Hackathon Problem Statements

Regd No: 25091A32J9, 25091A32G8

Problem Statement No: 4

Soil Health Classification System Develop a soil health card generator that classifies soil samples into fertility categories using clustering (K-means) on NPK, pH, and micronutrient datasets from government soil testing labs. The system should recommend suitable crops and fertilizer dosages based on the cluster the sample falls into. Use OOP to model different soil types as class hierarchies with inherited fertilization rules. The output should be a simple, vernacular-friendly report farmers can act on immediately.

Regd No: 25091A32E7, 25091A32H7

Problem Statement No: 20

Patient Queue Management System for Rural Clinics Build a queue management system for primary health centers that uses priority scheduling algorithms to balance walk-in patients, appointments, and emergency cases fairly and efficiently. Apply basic predictive analytics on historical footfall data to forecast peak hours and recommend staffing adjustments. Use OOP to model doctors, patients, and appointment slots with clear state management. This reduces long, unmanaged waiting times that discourage rural patients from seeking timely care.

Regd No: 25091A3243, 25091A3247

Problem Statement No: 57

Dropout Risk Prediction System Design a dropout risk prediction tool using classification models trained on attendance, academic performance, and socio-economic indicator datasets to flag at-risk students early. Generate an explainable risk report to guide teacher and counselor intervention. Use OOP to model student records and risk-factor categories for extensibility. This supports government initiatives aimed at reducing school dropout rates, particularly among girls and economically disadvantaged students.

Regd No: 25091A32B0, 24091A32C0

Problem Statement No: 118

Community Disaster-Preparedness Scoring System Build a community preparedness assessment tool using a weighted rule engine (OOP design) that scores neighborhoods or villages on disaster-readiness parameters (evacuation plan awareness, emergency kit availability, shelter proximity). Use the scores to target awareness campaigns where preparedness is lowest. This shifts disaster management from purely reactive to proactive community-level resilience building.

Regd No: 25091A3278, 25091A3269

Problem Statement No: 123

Ration Distribution Fraud Detector Build an anomaly detection system that analyzes beneficiary transaction patterns at ration shops using statistical methods to flag suspicious activity like duplicate claims, ghost beneficiaries, or unusual quantity patterns. Use OOP to model beneficiaries, ration shops, and transaction records. This helps plug leakages in the Public Distribution System that divert subsidized food from genuine beneficiaries.

Regd No: 25091A3248, 25091A3208

Problem Statement No: 128

Waste-to-Resource Matching System Create a matching platform connecting NGOs, waste generators, and recyclers using graph-based matching algorithms that pair surplus resources (food, materials) with organizations that can use them, based on location and need type. Use OOP to model donors, recipients, and resource categories. This reduces wastage while supporting NGOs working on food security and resource redistribution.